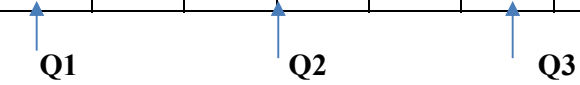


Western Mathematics Exams

Mathematics Standard

SOLUTIONS

Multiple Choice Worked Solutions		
No	Working	Answer
1	The constant k is equal to the gradient of the line Drawing a triangle on the line which passes through (0, 0) and (5, 16) Rise = 16 and Run = 5 $\text{so gradient} = \frac{\text{Rise}}{\text{Run}} = \frac{16}{5} = 3.2$	C
2	Although the numbers go in 20 amp increments the smallest markings are 5 amps apart, so the absolute error is half of this or 2.5 amps Lower boundary is $45 - 2.5 = 42.5$ amps Upper boundary is $45 + 2.5 = 47.5$ amps Actual amount lies between 42.5 amps and 47.5 amps	C
3	From graph 9% pa for 20 years earns \$1 800 on \$1000 investment If \$$X$ was \$5000 we would multiply 1800 by 5 i.e. $\frac{X}{1000}$ So on \$$X$ investment it is $\\$ \frac{1800X}{1000} = 1.8X$	B
4	$A \approx \frac{h}{2}(d_f + d_l)$ $h = 20$ $A = \frac{20}{2}(45 + 32) + \frac{20}{2}(32 + 40)$ $= 770 + 720$ $= 1\,490 \text{ m}^2$	B
5	42 hour week is 38 hours at normal rates and 4 hours at time and a half Equivalent normal hours = $38 + 4 \times 1.5 = 44$ hours Pay due = $44 \times 54 = \\$2\,376.00$	A
6	Because words are used to create categories it is Categorical, and because there is a hierarchy to the categories it is Ordinal	B
7	XR3 – 150km = 1.5 lots of 100 km rate = $\frac{12}{1.5} = 8$ litres per 100 km MY5 – 240 km = 2.4 lots of 100 km rate = $\frac{18}{2.4} = 7.5$ litres per 100 km KX2 – 200 km = 2.0 lots of 100 km rate = $\frac{15}{2.0} = 7.5$ litres per 100 km Both the KX2 and the MY5 have the lowest fuel consumption rates.	D

8	$\begin{aligned} P(\text{at least one number less than six}) &= P(\text{Not both numbers greater than five}) \\ &= 1 - P(\text{Both numbers greater than five}) \\ &= 1 - \frac{9}{64} \\ &= \frac{55}{64} \end{aligned}$	D																				
9	$\begin{aligned} g &= \frac{4}{9}f + 12 \\ g - 12 &= \frac{4}{9}f \\ 9(g - 12) &= 4f \qquad \text{OR} \qquad 9g - 108 = 4f \\ f &= \frac{9(g-12)}{4} \qquad \qquad \qquad f = \frac{9g-108}{4} \\ & \qquad \qquad \qquad f = \frac{9(g-12)}{4} \end{aligned}$	D																				
10	Find quartiles <table border="1"><tr><td>Tali</td><td>Pete</td><td>Fatma</td><td>Anna</td><td>Mia</td><td>Keith</td><td>Sue</td><td>Jack</td><td>Gloria</td><td>Mick</td></tr><tr><td>165</td><td>168</td><td>169</td><td>170</td><td>170</td><td>171</td><td>172</td><td>175</td><td>179</td><td>185</td></tr></table>  Q1 Q2 Q3 $IQR = 175 - 169 = 6$ Outliers are less than $169 - 1.5 \times 6 = 160$ (There are none less than 160) more than $175 + 1.5 \times 6 = 184$ (Only Mick is greater than 184)	Tali	Pete	Fatma	Anna	Mia	Keith	Sue	Jack	Gloria	Mick	165	168	169	170	170	171	172	175	179	185	A
Tali	Pete	Fatma	Anna	Mia	Keith	Sue	Jack	Gloria	Mick													
165	168	169	170	170	171	172	175	179	185													

**2020 Year 11 Final Examination
Mathematics Standard**

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 25 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☐ | B ☐ | C ☒ | D ☐ |
| 2. | A ☐ | B ☐ | C ☒ | D ☐ |
| 3. | A ☐ | B ☒ | C ☐ | D ☐ |
| 4. | A ☐ | B ☒ | C ☐ | D ☐ |
| 5. | A ☒ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☒ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☒ |
| 8. | A ☐ | B ☐ | C ☐ | D ☒ |
| 9. | A ☐ | B ☐ | C ☐ | D ☒ |
| 10. | A ☒ | B ☐ | C ☐ | D ☐ |

	Solution	Marks	Allocation of marks
Question 11			
	Dosage for children 1 – 12 years $= \frac{\text{age of child (in years)} \times \text{adult dosage}}{\text{age of child (in years)} + 12}$ $\text{Dosage} = \frac{9 \times 5}{9 + 12}$ $= \frac{45}{21}$ $= 2.142857$ $= 2.1 \text{ ml (nearest tenth of ml)}$	2	2 marks for correctly rounded correct answer 1 mark for incorrectly rounded correct answer or for correctly rounded incorrect answer
Question 12			
	Rounding to 2 significant figures gives 0.0000046 m In Scientific notation this is 4.6×10^{-6} m	2	2 marks for correct answer 1 mark if rounding or SN is correct only
Question 13			
	The shape is asymmetrical or not symmetrical and it is bi-modal.	2	2 marks for two correct statements or terms 1 mark for one correct statements or term
Question 14			
	$s = ut + \frac{at^2}{2}$ $210 = 5 \times 6 + \frac{a(6^2)}{2}$ $210 = 30 + \frac{36a}{2}$ $210 = 30 + 18a$ $180 = 18a$ $a = \frac{180}{18}$ $a = 10$	2	2 marks for correct answer 1 mark for correct substitution into formula with an attempt to solve equation or equivalent merit
Question 15			
(a)	A 15% loss leaves 85% of value Selling price to Mary = $0.85 \times 1200 = \\$1020$ Mary's profit = $1173 - 1020 = \\$153$ Mary's percentage profit = $\frac{153}{1020} \times 100 = 15\%$	2	2 marks for correct answer 1 mark for working with correct calculation of the loss or subsequent profit from incorrect loss

Question 16

$$\begin{aligned} \text{Area of end} &= \frac{\theta}{360} \times \pi r^2 \\ &= \frac{90}{360} \times \pi \times 40^2 \\ &= 1256.63706144 \\ \text{Volume} &= 1256.6371 \times 120 \\ &= 150\,796.452 \text{ cm}^3 \\ \text{Capacity} &= 150796.452 \div 1000 \\ &= 150.796 \text{ litres} \\ &= 151 \text{ litres nearest litre} \end{aligned}$$

2

2 marks for correct answer
 (Don't deduct marks for not rounding)

1 mark for an answer not converted to litres or with error in calculation using correct formula or equivalent merit

Question 17

Probability of meeting one of the nations

$$= \frac{9}{50} + \frac{1}{12} + \frac{3}{20} + \frac{1}{6}$$

$$= \frac{29}{50} = 0.58$$

$$\begin{aligned}\text{Expected Frequency} &= np \\ &= \frac{29}{50} \times 600 \\ &= 348\end{aligned}$$

Alternatively calculate the number for each nation separately

Nation	Bundjalung	Dharug	Kamilaroi	Wiradjuri
Number	108	50	90	100

$$\text{Total of four nations} = 108 + 50 + 90 + 100 = 348$$

2

2 marks for correct total or
for four individual totals

1 mark for finding total probability of four nations or for correct reasoning with a calculation error at some point

Question 18

$$y = 3x - 3$$

$$x + 2y = 8$$

$$y \text{ intercept} = -3 \quad (0, -3)$$

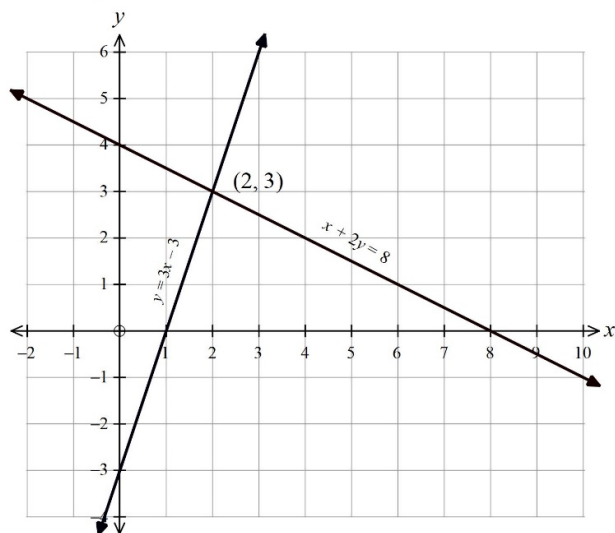
$$x = 0 \Rightarrow 2y = 8$$

$$\text{gradient} = 3 \Rightarrow (1, 0) \text{ or } (2, 3)$$

$$y = 4 \quad (0,4)$$

$$y = 0 \Rightarrow x = 8 \quad (8, 0)$$

Or can complete tables of values.

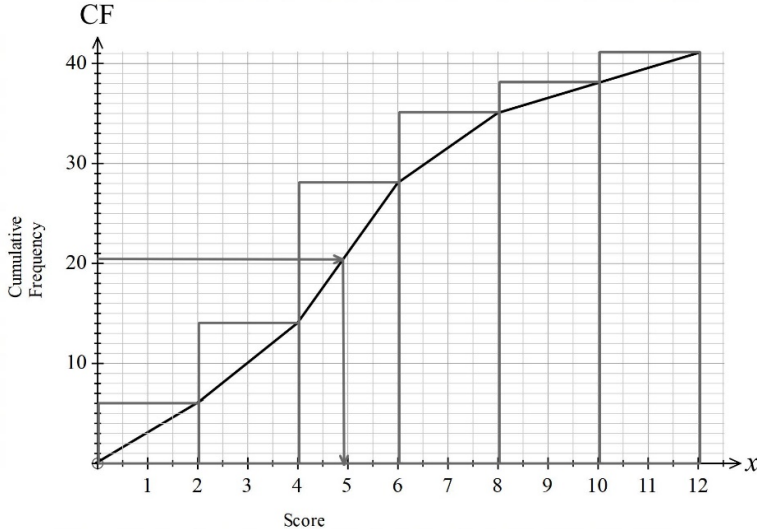


3

3 marks for correct graphs
and point of intersection

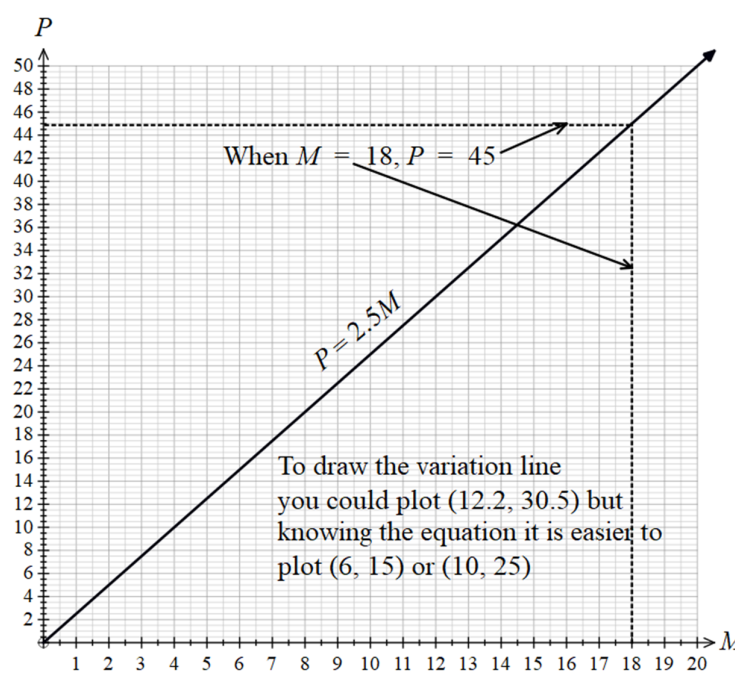
2 marks for one of the graphs correct and a point of intersection found correctly from a second incorrect graph

1 mark for working that shows some understanding of graphing techniques

Question 19						
(a)	Preparation Time	Class Centre	Frequency	Cumulative Frequency	2	2 marks for both columns correct 1 mark for one correct column completed correctly or for minor errors in both columns
	0 – 2	1	6	6		
	2 – 4	3	8	14		
	4 – 6	5	14	28		
	6 – 8	7	7	35		
	8 – 10	9	3	38		
	10 – 12	11	3	41		
(b) $\bar{x} = \frac{1 \times 6 + 3 \times 8 + 5 \times 14 + 7 \times 7 + 9 \times 3 + 11 \times 3}{6 + 8 + 14 + 7 + 3 + 3}$ $= \frac{209}{41}$ $= 5.09756098$ $= 5.1$ (one decimal place) Or by entering in <i>SD</i> mode on calculator					1	1 mark for correct answer
(c)  Median is around 4.9					2	2 marks for correct polygon and valid estimate of median from 4.8 – 5.0 1 mark for histogram instead of polygon or polygon without estimate of median
Question 20						
	Mass of one full carton in kg = $0.25 \times 30 + 1.2 = 8.7$ kg Mass of a loaded pallet = $8.7 \times 60 + 25 = 547$ kg Mass of four loaded pallets = 547×4 $= 2188$ kg = 2.188 tonnes $= 2.2$ tonnes (2 s f)				2	2 marks for correct answer 1 mark for correct reasoning with an error at some point

Question 21			
	Anzac Senior High Jillaroo Girls High Min age (Anzac) = 15.0 Median = 17 Median = 14 Min age (Jillaroo) = 11.5 Max age (Anzac) = 18.75 Max age (Jillaroo) = 19.0 Age (years) The median age for Anzac High is 3 years older than Jillaroo High (17 compared to 14) The ages are twice as widely spread at Jillaroo, (range of 7.5), compared to Anzac High (range of 3.75). Other measures that might be used are Interquartile range Anzac High (2.25) less than Jillaroo (3.75) Median of Anzac (17) is higher than upper quartile of Jillaroo (16.5) Min value of Anzac (15) is higher than median of Jillaroo (14)	2	2 marks for any valid comparison that included the medians and mentions that Anzac are higher and more compressed than Jillaroo 1 mark for a comparison that only mentions comparison of central tendency or of spread, not both
Question 22			
	Cycling = $\frac{40}{60} \times 55 \times 23.0 = 843.3$ Swimming = $\frac{10}{60} \times 55 \times 22.1 = 202.583$ Jogging = $\frac{70}{60} \times 55 \times 34.1 = 2188.083$ Total burned = 3234.0 kilojoules	2	2 marks for correct answer 1 mark for some correct and relevant calculations
Question 23			
(a)	Stamp duty = $(54200 - 45000) \times 0.05 + 1350$ = $9200 \times 0.05 + 1350$ = $460 + 1350$ = \$1 810	2	2 marks for correct answer 1 mark for some correct and relevant calculations
(b)	Cost = $1810 + 453 + 54200$ = \$56 463.00	1	1 mark for correct equation
Question 24			
(a)	CD is a vertical line so has equation $x = 6$	1	1 mark for correct equation
(b)	BD has gradient = $\frac{2 - -4}{3 - 6} = \frac{6}{-3} = -2$ y intercept = 8 Equation $y = -2x + 8$	1	1 mark for correct equation
(c)	BC has gradient = $\frac{4 - 2}{6 - 3} = \frac{2}{3}$ y intercept = 0 Equation $y = \frac{2}{3}x$	1	1 mark for correct equation

Question 25			
	Calculate the distance PQ $QU = 39 - 15$ $= 24 \text{ m}$ $PQ^2 = 10^2 + 24^2$ $= 676$ $PQ = \sqrt{676}$ $= 26$ Perimeter $= 28 \times 2 + 39 + 15 + 10 + 26$ $= 146 \text{ m}$	2	2 marks for correct answer 1 mark for perimeter found correctly from incorrect value of PQ or equivalent merit
Question 26			
(a)	$P(\text{Blue Black}) = 0.4 \times 0.3 = 0.12$	1	1 mark for correct answer
(b)	$P(\text{Red Orange}) = 0.3 \times 0.2 = 0.06$ $P(\text{Not Red Orange}) = 1 - 0.06 = 0.94$	1	1 mark for correct answer
(c)	$P(\text{Red Black}) = 0.3 \times 0.3 = 0.09$ $P(\text{Yellow Green}) = 0.1 \times 0.5 = 0.05$ $P(\text{Red Black or Yellow Green}) = 0.09 + 0.05 = 0.14$	1	1 mark for correct answer
Question 27			
	Number of books sold $= 8 \times 940 = 7\,520$ Amount earned by sales $= 7\,520 \times 18.95 = \$142\,504$ Royalties $= 142\,504 \times 0.0625 = \$8\,906.50$ Amount earned $= 8\,906.5 + 12\,500$ $= \$21\,406.50$	2	2 marks for correct answer 1 mark for some correct and relevant working
Question 28			
(a)	Swim distance travelled $= 3 \times 500$ $= 1\,500 \text{ m}$ $= 1.5 \text{ km}$ Swim time taken $= 30 \text{ min} = 0.5 \text{ hr}$ $\text{Speed} = \frac{D}{T} = \frac{1.5}{0.5} = 3 \text{ km/h}$	2	2 marks for correct answer 1 mark for correct use of formula without correct units or similar merit
(b)	Bike distance travelled $= 40 \text{ km}$ Bike Speed $= 35 \text{ km/h}$ $T = \frac{D}{S}$ $= \frac{40}{35}$ $= 1.14285714 \text{ hours}$ $= 1 \text{ hour } 9 \text{ min (nearest minute)}$	2	2 marks for correct answer in hours and minutes (Accept 8 or 9 minutes – i.e. don't penalise rounding as long as time is found correctly) 1 mark for correct use of formula without changing to hours and minutes or similar merit

Question 29			
	Original value of first 3 terminals = $980 \times 3 = \\$2940$ Annual loss in value = $2940 \times 0.16 = \\$470.40$ Current value 1st 3 terminals use : $S = V_0 - Dn$ $S = 2940 - 470.4 \times 3 = \\$1\,528.80$ Original value of next 3 terminals = $850 \times 3 = \\$2550$ Annual loss in value = $2550 \times 0.16 = \\$408.00$ Current value 2nd 3 terminals = $2550 - 408 = \\$2\,142.00$ Current value of all terminals = $\\$1\,528.80 + \\$2\,142.00 = \\$3\,670.80$	2	2 marks for correct answer 1 mark for use of correct formula with some correct and relevant calculations
Question 30			
	 To draw the variation line you could plot (12.2, 30.5) but knowing the equation it is easier to plot (6, 15) or (10, 25) $P = kM$ $30.5 = 12.2k$ $k = \frac{30.5}{12.2} = 2.5$ $P = 2.5M$ Otherwise method When $M = 18$ $P = 2.5M$ $= 2.5 \times 18$ $= 45$	4	4 marks for correct equation, correct graph and the correct value $P = 45$. 3 marks for correct graph and correct working with a minor error in finding P or equivalent merit 2 marks for incorrect graph used to find a value of P correctly or equivalent merit 1 mark for some relevant and correct working or reasoning.
Question 31			
(a)	Taxable income = $67626 + 420 + 34000 - 12500$ = $\\$89\,546$	1	1 mark for correct answer
(b)	Tax Due = $(89546 - 37000) \times 0.325 + 3572$ = $52546 \times 0.325 + 3572$ = $17077.45 + 3572$ = $\\$20\,649.45$ PAYG Tax paid = $520.50 \times 26 = \\$13\,533.00$ Tax Bill = $20649.45 - 13533.00 = \\$7\,116.45$	3	3 marks for correct answer 2 marks for correctly finding the tax paid or equivalent merit 1 mark for correctly finding the tax due or equivalent merit

Question 32					
(a)	Total of Frequencies = $24 + 16 + 28 + 12 = 80$		1	1 mark for correct column	
	Material Used	Frequency			Relative Frequency
	Wooden	24			$\frac{24}{80} = 0.3$
	Metal	16			$\frac{16}{80} = 0.2$
	Ceramic	28			$\frac{28}{80} = 0.35$
	Fabric	12			$\frac{12}{80} = 0.15$
(b)	$P(\text{Fabric}) = 0.15$ $P(\text{Not Fabric}) = 1 - 0.15 = 0.85$		1	1 mark for correct answer	
(c)	Number of Not Fabric = 68 Number of metal = 16 $P(\text{Metal given not fabric}) = \frac{16}{68} = \frac{4}{17} \approx 0.23529$		1	1 mark for correct answer	
Question 33					
	As both are the same day, so the first must be in the morning so it is 10: 35 in 24 hour time. Want the time elapsed from 10: 35 to 22: 15. From 10: 35 to 22: 35 is 12 hours, but to 22: 15 is 20 min less. So time taken is 12 hours – 20 min = 11 hours and 40 minutes OR From 10: 35 to 11: 00 is 25 min From 11: 00 to 22: 00 is 11 hours From 22: 00 to 22: 15 is 15 min 11 hours + 25 min + 15 min = 11 hours and 40 minutes		2	2 marks for correct answer 1 mark for answer with correct reasoning, but a calculation error or equivalent merit	
Question 34					
	Both the cylinder and the hemisphere have a radius of 20 cm. Surface area of an open cylinder = $2 \pi r h$ Surface area of a hemisphere = $4 \pi r^2 \div 2 = 2 \pi r^2$ Surface area of dome = $2 \pi r h + 2 \pi r^2$ = $2 \times \pi \times 20 \times 25 + 2 \times \pi \times 20^2$ = $3141.59265359 + 2513.27412287$ = 5654.86677646 = 5655 cm^2 (nearest cm^2) Accept exact answer = $1\,800\pi \text{ cm}^2$		2	2 marks for correct answer 1 mark for either of the partial areas found correctly or equivalent merit	

Question 35			
	Sitiveni is 2 hours ahead of Canberra so 16:00 on Mon Kirsty is 10 hours behind Canberra, so 04:00 on Mon Joan is 7 hours behind Kirsty so this takes you past midnight and back to Sunday, so her time is 21:00 on Sunday Edouard is a further 3 hours back so he is 18:00 on Sunday In Summary Sitiveni -16:00 on Monday Kirsty - 04:00 on Monday Joan - 21:00 on Sunday Edouard - 18:00 on Sunday	4	4 marks for all four correct times 3 marks for 3 correct times or 2 correct with one error following on correctly from a prior error 2 marks for 2 correct with two errors 1 mark for 1 correct
Question 36			
(a)	Amount A is based on his usage, and amount B is a daily usage fee which is fixed by the company. If he wishes to reduce his bill he could reduce amount A by looking at ways to lower his usage, for example setting any thermostats a degree lower or taking shorter showers. He could also contact his supplier and try to renegotiate his rates to lower either A or B or both.	1	1 mark for any reasonable explanation, that states that he only has control over A
(b)	Amount of Jan March gas bill = $\\$158.07 \times 0.7$ = \$110.65 Total of gas bills = $110.65 \times 2 + 158.07 \times 2$ = \$537.44 Total of all current bills = $537.44 + 910.25 \times 2$ = $537.44 + 1820.50$ = \$2357.94 Anticipated increased bills in 2021 = 2357.94×1.12 = \$2640.8928 Amount each fortnight = $2640.8928 \div 26$ = \$101.57	3	3 marks for correct answer 2 marks for basically correct reasoning with minor error(s) 1 mark for 2 or more correct and relevant calculations leading to an incorrect answer or equivalent merit